

Renal Reference — AKI Pattern Table

Clean “**pattern table**”

Finding	Prerenal AKI	ATN / Intrinsic AKI	Postrenal AKI
Main problem	↓ renal perfusion	Tubular/parenchymal injury	Urine obstruction
Kidney structure	Intact	Damaged	Initially intact, damaged if prolonged
BUN/Cr	>20:1	10–15:1	Variable
FeNa	<1%	>1%	Variable
FeUrea	<35%	>50%	Variable
Urine Na	<20 mEq/L	>40 mEq/L	Variable
Urine osm	>500	<350	Variable
Urine sediment	Bland, hyaline casts	Muddy brown granular casts, RTEC casts	Crystals, hematuria, or bland
Response to fluids	Improves	Does not rapidly improve	Improves only after relieving obstruction

1. Prerenal column

WHAT YOU SEE

Column header 'Prerenal AKI'

WHAT IT ENCODES

The hypoperfusion / 'conserving' pattern. A structurally intact nephron under RAAS + ADH activation avidly reabsorbs Na and water: FENa <1%, urine Na <20, urine osm >500, BUN:Cr >20, bland/hyaline-cast sediment. Causes: volume loss, HF, cirrhosis, sepsis, NSAIDs/ACE-I. Fully reversible if perfusion is restored before ischemic ATN develops.

BOARD TRAP

WRONG answer is treating cardiorenal or hepatorenal 'prerenal' physiology with crystalloid boluses. These patients are total-body overloaded with low EFFECTIVE volume. Discriminator: HF cardiorenal AKI needs decongestion/improved output; HRS needs albumin + vasoconstrictors (terlipressin/midodrine-octreotide) — saline worsens both despite the prerenal indices.

2. ATN / Intrinsic column

WHAT YOU SEE

Column header 'ATN / Intrinsic AKI'

WHAT IT ENCODES

The tubular-injury / 'wasting' pattern. Damaged tubules cannot reabsorb Na or concentrate urine: FENa >2%, urine Na >40, urine osm <350 (isosthenuric), BUN:Cr 10-15, MUDDY-BROWN granular and RTEC casts. Causes: ischemia, sepsis, nephrotoxins (aminoglycosides, contrast, cisplatin), pigments (rhabdo/hemolysis). Phases: initiation → maintenance (oliguric) → recovery (polyuric).

BOARD TRAP

WRONG answer is assuming all ATN starts with a high FENa. Early sepsis-, contrast-, and pigment-associated ATN can show FENa <1% before frank necrosis. Discriminator: the muddy-brown granular casts are more specific for ATN than the FENa — follow the sediment and Cr trend when the index is misleadingly low.

3. Postrenal column

WHAT YOU SEE

Column header 'Postrenal AKI'

WHAT IT ENCODES

The obstruction pattern. Indices are VARIABLE: early on the kidney behaves prerenal-like (low FENa, concentrated urine); with prolonged obstruction it shifts to an ATN-like wasting pattern (high FENa, isosthenuria). Requires bilateral obstruction (or a solitary functioning kidney). Diagnosed by bladder scan + renal ultrasound; improves only after relieving the obstruction.

BOARD TRAP

WRONG answer is ruling out postrenal AKI because the FENa is low or the ultrasound shows no hydronephrosis. Early obstruction gives low FENa, and dehydrated/encased obstruction can lack hydronephrosis. Discriminator: confirm with a post-void residual/bladder scan and, if suspicion is high, CT — relief of obstruction (Foley/stent/nephrostomy) is both diagnostic and therapeutic.

4. Main problem

WHAT YOU SEE

Row 'Main problem'

WHAT IT ENCODES

The core lesion defining each column: prerenal = ↓renal PERFUSION (intact nephron, hypoperfused); ATN/intrinsic = tubular/parenchymal INJURY (necrotic tubular cells); postrenal = mechanical OBSTRUCTION to urine outflow. Identifying which 'problem' is present dictates therapy — restore perfusion, support and remove toxins, or relieve obstruction, respectively.

BOARD TRAP

WRONG answer is applying the same treatment to all three (e.g., fluids for everything). Discriminator: fluids help prerenal, do little for established ATN (and harm if oliguric/overloaded), and are irrelevant to postrenal, which needs decompression — match the intervention to the underlying problem.

5. Kidney structure

WHAT YOU SEE

Row 'Kidney structure'

WHAT IT ENCODES

Prerenal: structurally INTACT (functional, fully reversible). ATN: tubular cells DAMAGED/necrotic (structural injury, recovery over days-weeks). Postrenal: initially intact but becomes damaged if obstruction is PROLONGED (irreversible after weeks of high back-pressure). This explains why prerenal reverses fast while delayed relief of obstruction may not fully recover.

BOARD TRAP

WRONG answer is assuming relieved obstruction always restores baseline function. Prolonged obstruction (weeks) causes permanent tubulointerstitial damage and incomplete recovery. Discriminator: the longer the kidney has been obstructed, the lower the chance of full recovery — relieve obstruction PROMPTLY and counsel about possible residual CKD.

6. BUN/Cr

WHAT YOU SEE

Row 'BUN/Cr'

WHAT IT ENCODES

Prerenal >20:1 — low tubular flow drives urea reabsorption (with Na/water) while creatinine is not reabsorbed. ATN 10-15:1 — impaired tubular reabsorption normalizes the ratio. Postrenal variable depending on duration/phase.

BOARD TRAP

WRONG answer is calling any BUN:Cr >20 prerenal. GI bleeding, high protein/TPN, steroids, and tetracyclines raise BUN extrarenally. Discriminator: corroborate with FENa <1% and bland sediment; a melena/hematemesis history explains a high BUN:Cr without true prerenal physiology.

7. FeNa

WHAT YOU SEE

Row 'FeNa'

WHAT IT ENCODES

FENa = (urine Na × plasma Cr)/(plasma Na × urine Cr) × 100. Prerenal <1% (Na avidly conserved); ATN >2% (Na wasted by injured tubules); postrenal variable. It is the most-tested discriminator on boards — but is reliable only in oliguric AKI off diuretics.

BOARD TRAP

FeNa <1% always = prerenal — this is the WRONG answer. Low-FENa EXCEPTIONS (intrinsic AKI): early sepsis, contrast nephropathy, pigment nephropathy (rhabdo/hemolysis), early/acute GN, and HRS. Discriminator: when low FENa coexists with muddy-brown casts, elevated CK, recent contrast, or RBC casts, the lesion is intrinsic — pair FENa with the sediment.

8. FeUrea

WHAT YOU SEE

Row 'FeUrea'

WHAT IT ENCODES

Fractional excretion of urea: prerenal <35%, ATN >50%. Because urea handling is largely independent of the distal Na transport that diuretics block, FeUrea is the preferred index when the patient is on loop/thiazide diuretics (which artificially raise FENa).

BOARD TRAP

WRONG answer is using FENa to call ATN in a patient actively on furosemide. Diuretics force natriuresis and push FENa >1% even in true prerenal states. Discriminator: switch to FeUrea — a value <35% indicates prerenal AKI despite a misleadingly high FENa under diuresis.

9. Urine Na

WHAT YOU SEE

Row 'Urine Na'

WHAT IT ENCODES

Prerenal urine Na <20 mEq/L (RAAS/ADH-driven sodium conservation); ATN urine Na >40 mEq/L (injured tubules cannot reabsorb Na). It tracks with FENa but is a cruder, single-sample measure subject to the same exceptions and diuretic effects.

BOARD TRAP

WRONG answer is interpreting urine Na in isolation right after a diuretic dose. Loop/thiazide diuretics raise urine Na regardless of volume status. Discriminator: time the sample away from diuretics, or rely on FeUrea; and remember a low urine Na is one of the low-FENa exception scenarios (sepsis/contrast/pigment) when intrinsic injury is present.

10. Urine osm

WHAT YOU SEE

Row 'Urine osm'

WHAT IT ENCODES

Prerenal urine osm >500 mOsm/kg — the intact, ADH-responsive nephron concentrates urine. ATN urine osm <350 (isosthenuric, fixed near plasma ~300) — injured tubules lose concentrating ability. A high urine osm with oliguria favors a functional (prerenal) kidney.

BOARD TRAP

WRONG answer is expecting high urine osm to persist into established ATN or after diuretics/osmotic agents. Loss of concentrating ability (isosthenuria) is a hallmark of tubular injury. Discriminator: urine osm <350 with muddy-brown casts confirms ATN; a urine osm >500 in oliguric AKI points to a still-functional prerenal nephron.

11. Urine sediment

WHAT YOU SEE

Row 'Urine sediment'

WHAT IT ENCODES

The most specific localizing test. Prerenal: BLAND, occasional hyaline casts. ATN: MUDDY-BROWN granular + renal tubular epithelial cell casts. Postrenal: crystals, hematuria, or bland. (Adjacent intrinsic patterns for contrast: RBC casts/dysmorphic RBCs = GN; WBC casts + eosinophiluria = AIN.)

BOARD TRAP

WRONG answer is overriding the sediment with a conflicting FENa. The sediment is MORE specific for localizing the lesion than the indices. Discriminator: muddy-brown granular casts diagnose ATN even if FENa is <1% (early sepsis/contrast/pigment exception); hyaline-only/bland sediment supports prerenal even if a diuretic pushed FENa above 1%.

12. Response to fluids

WHAT YOU SEE

Row 'Response to fluids'

WHAT IT ENCODES

Prerenal IMPROVES with restored perfusion/volume (creatinine falls within 24-72h). ATN does NOT rapidly improve with fluids — it recovers over days to weeks of supportive care. Postrenal improves only after the obstruction is RELIEVED (Foley/stent/nephrostomy), not with fluids. A fluid challenge is therefore both diagnostic and therapeutic in prerenal AKI.

BOARD TRAP

ATN responds to a fluid bolus — this is the WRONG answer. Once tubular necrosis is established, fluids do not reverse it and excess fluid causes pulmonary edema in an oliguric patient. Discriminator: a creatinine that falls promptly after volume = prerenal; failure to improve (with muddy-brown casts, FENa >2%) = ATN — stop fluids, avoid nephrotoxins, and provide supportive/RRT care as needed.